



CHURN PREDICTION IN DODGE THE MUD GAME

Analysis of Customer Data

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1. Problem definition

Dodge the Mud is a free mobile game in which players try to avoid falling mud. The dataset contains gameplay sessions from January 2015 to May 2016, consisting of 163,929 rows and three variables: device, score, and time. The time variable indicates Unix Epoch timestamps.

In the context of mobile games, churn refers to a player who permanently stops playing (Runge et al., 2014). To construct reliable labels for churn and non-churn, two-time windows are required: the Observation Period (OP) and the Churn Prediction Period (CP). The OP begins at the player's first recorded and captures early engagement behavior, while the CP follows the OP and determines churn status. Players with at least one session in the CP are considered non-churners; otherwise, they are labeled churners. These OP/CP windows are the foundation to create ground-truth labels.

The main purpose of this project is to build a machine pipeline for binary classification task to classify the churner and non-churner by (1) engineering features, (2) handling extreme class imbalance, (3) training three Machine Learning (ML) models, (4) hyperparameter tuning and (5) comparing models and interpreting predictions through SHAP analysis.

2. Methods

To achieve the above goal, the following methodologies are applied as shown in Figure 1:

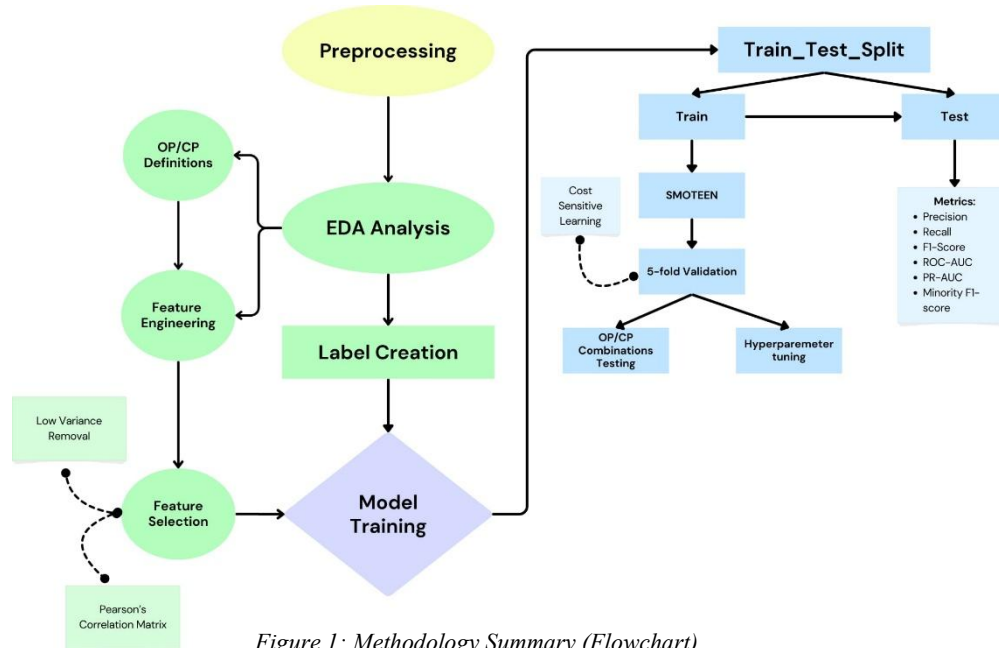


Figure 1: Methodology Summary (Flowchart)

2.1 EDA and Preprocessing Data

First, Unix timestamps are converted to datetime formats and split into components like months, day, time for EDA. Only nine missing device values are identified and removed, because it's only a small portion of the dataset. In EDA stage, we focus on 3 main patterns: Player retention and lifetime patterns (retention curve, player lifetime distribution), session behavior (inter-session intervals, session frequency and peak activity hours), and score trends (learning curve, score distribution). These analyses help to understand the data and decide reasonable OP/CP windows by revealing typical player lifetimes and engagement patterns.

2.2 Churn Label Creation

Because churn labels depend on how OP/CP windows are defined, choosing suitable OP/CP windows is a foundation for this project. Too short an OP risks misclassifying non-churners who are slow starters, while too short a CP may mislabel players who simply take breaks. Based on EDA, OP/CP windows were initially chosen by analyzing player retention and return-rate statistics: the OP was set to day 5, reflecting a sharp drop in retention, and the CP was chosen using return-rate statistics after the OP window to capture typical player return behavior. Moreover, after training models, four OP/CP combinations were evaluated to verify robustness and confirm the final configuration.

2.3 Feature Engineering

Feature engineering focuses on capturing early player behavior during the OP. With only 3 raw variables (device, score, time), careful feature engineering is critical. Too many features may cause noise in the model training, while few features cannot capture all the pattern. Therefore, we create a balanced feature set including 33 features. Both basic and advanced behavioral features are created, divided into different groups: score metrics, session pattern, temporal behavior, interaction features. Afterwards, feature selection is applied in 2 steps: low variance removal (threshold 1%) to eliminate non-informative features and pairwise correlation filtering ($\text{cor} > 0.9$) using Random Forest feature importance to retain most predictive features.

2.4 Class Imbalance Handle and Models training

A major challenge of this project is severe class imbalance. For current chosen OP = 5 and CP = 10, churners account for 92%. Applying sampling method to the whole dataset like SMOTE would likely generate excessive, unrealistic synthetic samples. After trying different imbalance handling techniques, SMOTEENN, a hybrid approach combining SMOTE with Edited Nearest Neighbors, is chosen to address the limitations of SMOTE by removing noisy samples after oversampling (Husain et al. 2025). However, SMOTEENN is applied to 50% of the dataset to mitigate potential distortion. The resulting distribution on train dataset was more manageable (66% churn, 33% non-churn).

After SMOTEENN, data was split into training and test sets, and models were evaluated using five-fold cross-validation. Three models were selected for training:

- **Random Forest:** A Random Forest is an ensemble of decision trees where each tree is trained on a bootstrapped sample of the data, and predictions are made through majority voting across trees (Breiman, 2001)
- **XGBoost:** a gradient boosting framework that constructs trees sequentially, with each tree trained to reduce the residual errors left by the previous ensemble (Chen & Guestrin, 2016).
- **CatBoost:** a gradient boosting algorithm designed to handle categorical features efficiently and reduce overfitting through ordered boosting and target statistics, leading to strong performance with minimal preprocessing (Dorogush, Ershov & Gulin, 2018).

These models were chosen because churn prediction typically relies on subtle behavioral patterns, and tree-based ensembles are well suited for capturing non-linear relationships without demanding extensive preprocessing. Given the severe class imbalance, cost-sensitive learning is incorporated to penalize misclassifications of the minority class more heavily (Araf et al., 2024). By assigning higher costs to false negatives, the models were encouraged to focus on correctly identifying non-churners.

2.5 Hyperparameter Tuning

To optimize model performance, each model undergoes hyperparameter tuning using Optuna, a state-of-the-art framework for efficient hyperparameter search. The objective functions in Optuna maximize the F1 score using stratified cross-validation to maintain class distributions. After obtaining the best hyperparameters, threshold optimization is performed to maximize the minority-class F1 score, explicitly addressing class imbalance. This ensures the final models are both sensitive to the minority class and robust overall.

2.6 Evaluation, Error Analysis, and SHAP Analysis

Model evaluation used metrics including accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. Despite satisfactory overall performance, the minority class (non-churners) remained challenging, so we focus more on accessing minority-class precision, recall, and F1-score. SHAP analysis on the best model provided interpretable insights into feature importance and churn drivers, combining predictive assessment with actionable understanding for business decisions.

3. Results

3.1 Model Performance and Comparison

Overall, the evaluation of three ensemble learning models, Random Forest, XGBoost, and CatBoost, consistently strong performance across standard metrics, with subtle distinctions among them (Figure 2). Error analysis reveals that most misclassifications occur in borderline cases, particularly players with minimal or inconsistent activity during the CP, highlighting inherent challenges in predicting low-activity churners.

Random Forest achieved an accuracy of 0.921, precision of 0.950, recall of 0.966, and F1-score of 0.958. Its ROC-AUC (0.781) and PR-AUC (0.974) indicate strong overall discriminative ability and precision-recall balance. It performs well in identifying true positives but shows moderate performance on minority class (Minority F1 = 0.408), suggesting limited sensitivity to minority class.

XGBoost shows slightly lower overall accuracy (0.919) but comparable precision (0.950) and recall (0.964), resulting in an F1-score of 0.957. While its PR-AUC is lower than Random Forest (0.969 vs. 0.974), XGBoost exhibits slightly higher minority-class recall than Random Forest (0.368 vs. 0.365), indicating it is better at capturing rare churn events but still with moderate effectiveness.

CatBoost delivers similar overall metrics (accuracy = 0.915, precision = 0.953, recall = 0.955, F1 = 0.954), with the highest ROC-AUC (0.789) among the three models. Importantly, CatBoost demonstrates the best balance for minority-class performance, achieving the highest minority recall (0.420) and F1 (0.424). Although its overall accuracy is slightly lower than the others, its sensitivity to the minority class and consistent cross-validation performance (CV ROC-AUC 0.974 vs. test ROC-AUC 0.789, gap = 0.184) make it the most reliable choice for this problem.

The comparison across models shows that no single model dominates every metric. Random Forest excels in overall F1 and PR-AUC, CatBoost leads in ROC-AUC and minority-class F1, and XGBoost provides a compromise in recall. These results highlight a trade-off between maximizing overall performance and addressing class imbalance, which is crucial in predicting churn for imbalanced data.

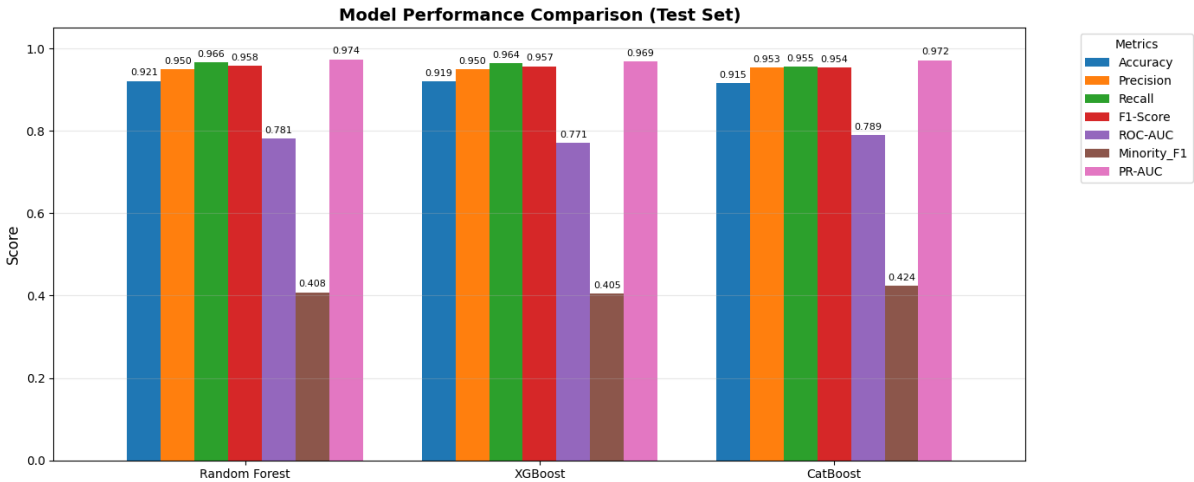


Figure 2: Model Comparison

3.3 Best Parameters

The optimal CatBoost model is identified through hyperparameter tuning via Optuna. The best trial yield: 'iterations': 534, 'depth': 8, 'learning_rate': 0.199, 'l2_leaf_reg': 5.395, 'border_count': 46. These parameters balance model complexity and learning rate, ensuring both strong predictive power and stable training.

3.4 Test OP/CP Combination

To confirm the optimal OP and CP windows, four suitable configurations are evaluated using stratified five-fold cross-validation with cost-sensitive learning. While the combination (**OP = 7, CP = 7**) achieved the highest overall metrics like ROC-AUC, F1 and PR-AUC, its performance on the minority class is notably poor (Minority F1 = 0.739 for XGB). Therefore, the configuration (**OP = 5, CP = 10**) was chosen, as it provides a balanced solution, maintaining competitive overall metrics, as visualized in Figure 3. Although not the absolute best across all metrics, it provides consistent performance while mitigating class imbalance, making it the most practical choice. This highlights a key trade-off: maximizing overall performance can come at the expense of accurately capturing rare or low-frequency churners.

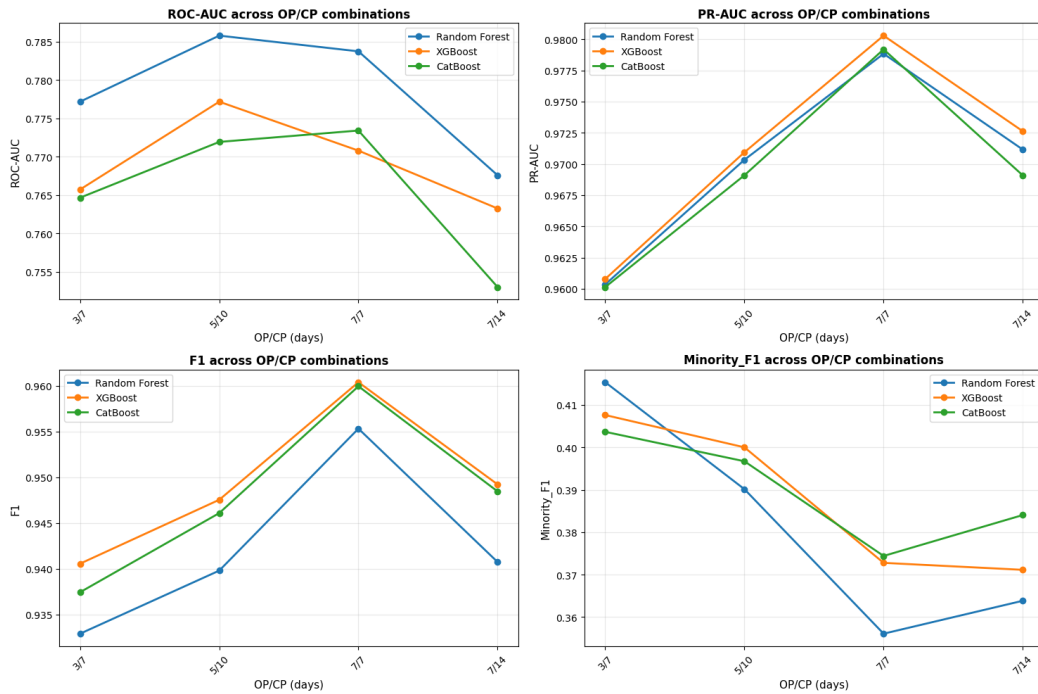


Figure 3 OP/CP Combination Comparison Across Models

3.5 SHAP Analysis

Aligning with the above findings, SHAP analysis is conducted on CatBoost – most performative model. Figures 4–5 show the top 10 features' SHAP distributions and rankings. Temporal features, particularly weekend_ratio, night_play_ratio, and mean_interval_hours, are the most influential based on their high SHAP values and wide dispersion. Low values of weekend_ratio and mean_interval_hours push predictions upward, whereas night_play_ratio has the opposite effect. Other temporal indicators, like avg_hour_of_day and hour_entropy, show consistent directional influence, with later-day activity increasing predicted risk.

Recency and behavioral diversity features (recency_days and score_entropy) exhibit contrasting effects: recent activity raises predicted churn risk, while higher behavioral entropy lowers it. Engagement-volume and consistency features, such as total_sessions, session_span_days, and score_cv, show more compact distributions, indicating stable influence; the first two follow positive trends, whereas score_cv is inverse. Overall, these patterns reveal that the model's predictions are

shaped by a combination of temporal, recency, diversity, and engagement signals, underscoring the multifaceted behavioral drivers of churn.

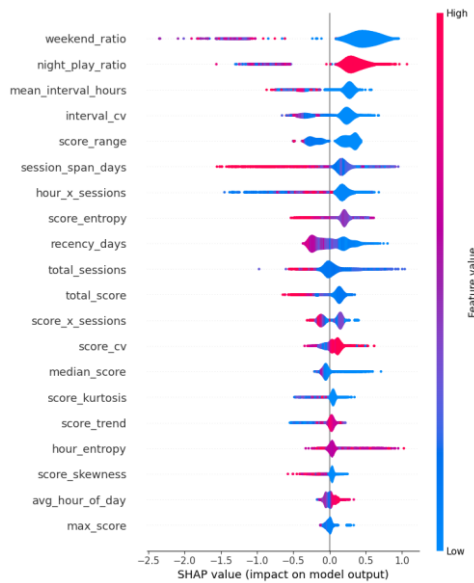


Figure 4: SHAP Beeswarm

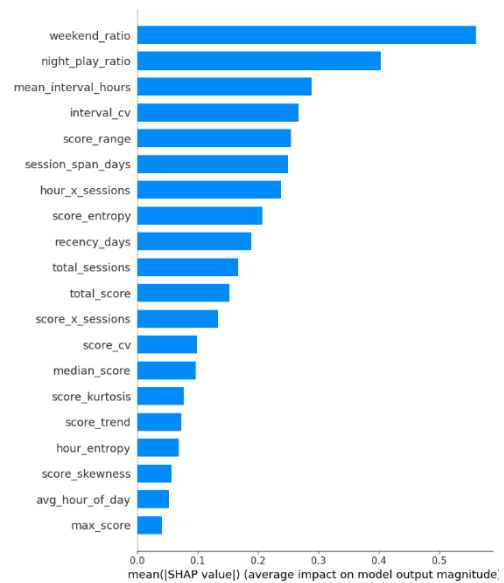


Figure 1: SHAP Importance Ranking

4. Discussion

The results demonstrate that early player behavior in *Dodge the Mud* provides a solid basis for predicting long-term churn. Among the ensemble methods, CatBoost emerged as the best-performing model, likely due to its strength in handling complex interactions and categorical features. Random Forest and XGBoost also performed strongly, indicating that tree-based ensemble approaches are generally well-suited to this problem.

A key methodological challenge was addressing extreme class imbalance, which can bias training toward the majority class and reduce minority-class predictive performance. Several approaches were tested, including SMOTE and SMOTETomek, but a hybrid approach combining SMOTEEN with cost-sensitive learning proved most effective. This highlights the trade-off between improving minority-class sensitivity and maintaining overall model stability. Despite these efforts, minority-class F1 scores remain moderate, reflecting the intrinsic difficulty of predicting churn among low-activity players.

Another important trade-off involved the selection of $OP = 5$ and $CP = 10$. While configurations with larger observation periods (e.g., $OP = 7$, $CP = 7$) achieved higher overall metrics, they performed poorly on the minority class. Conversely, shorter windows reduced dataset coverage and model robustness. The chosen configuration balances overall predictive performance, minority-class sensitivity, and label reliability, illustrating the importance of carefully tuning temporal windows for actionable predictions. This experiment highlights that varying the window can substantially change the results and model interpretability (Bugajev et al., 2022) and that beyond a certain length of memory window (history).

Feature engineering also played a critical role. While many features were initially created, only a compact subset, primarily temporal features such as `weekend_ratio`, `night_play_ratio`, and `mean_interval_hours` had substantial predictive power. This suggests that early player behavior is relatively simple yet highly informative and that including excessive or redundant features offers diminishing returns. More recently, Peters, et al. (2023) extended this insight by showing that temporal-behavioral patterns capture non-redundant variance beyond static usage features, thereby enhancing the performance of deep LSTM neural networks.

Regarding the limitations, despite the use of hybrid imbalance-handling methods and hyperparameter tuning, the models still struggle with the minority class, indicating that further improvements could require additional contextual or longitudinal features. Additionally, while OP/CP tuning mitigates some trade-offs, it introduces a degree of arbitrariness in defining observation and churn periods, which may affect generalizability to other games or domains.

5. Conclusion

This project develops a predictive model for identifying player churn in *Dodge the Mud*, with CatBoost achieving the best overall performance. Evaluating multiple OP/CP configurations, mitigating class imbalance, and engineering relevant features are key to building a robust model. Early player activity, particularly within the first five days, proves highly informative for predicting future churn. Temporal, recency, and behavioral diversity features emerged as the most influential, offering actionable insights for targeted retention strategies. Overall, the study highlights the trade-offs between overall performance, minority-class sensitivity, and feature complexity, emphasizing the value of interpretable machine learning for guiding both model development and business decisions.

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Declarations statement

- Code (Preprocessing, EDA, Model Training, OP/CP Comparison): Le Anh Thu Pham
- Code (Hyperparameter Tuning, Model Comparison, SHAP Analysis): Lam Minh Triet Thien
- Writing (Problem Statement, Discussion, Conclusion): Lam Minh Triet Thien
- Writing (Methodology, Results, Review and Final Edits): Le Anh Thu Pham

Use of Generative AI Tools

- Claude: Used to improve data visualizations for EDA, clean and optimize code, identify and fix bugs, and generate clearer plots for model comparison.
- ChatGPT: Consulted to explore methods to handle class imbalance, and help to evaluate different approaches, as well as suggestions for hyperparameter tuning strategies.